Lesson 29

Scientists Find ‘Chorus Waves’ in Unexpected Part of Space

International researchers recently released a study on bursts of energy in an unexpected area of space.

Scientists call the energetic activity “chorus waves” because they are said to move at the same frequency as sounds humans can hear. When the energy is turned into audio signals, the waves sound like birds chirping.

Researchers have known about chorus waves in space for many years. But the latest research has found the waves exist 100,000 kilometers from Earth. Chorus waves have never before been measured at such a big distance from a planet.

In an email, study writer Chengming Liu from Beihang University in Beijing wrote: “They are one of the strongest and most significant waves in space.”

Allison Jaynes is a space physicist at the University of Iowa who was not involved with the study. Speaking about the recent study, Jaynes said, “That opens up a lot of new questions about the physics that could be possible in this area.”

The newfound chorus waves were detected in an area where Earth's magnetic field is stretched out. Scientists said they did not expect to find the waves there. The discovery raises new questions about how the chirping waves form.

Jaynes added, “We definitely need to find more of these events.”

Scientists still are not sure how the waves develop. But they think Earth’s magnetic field may have something to do with it.

For many years, radio antennas have picked up chorus waves. This includes receivers at an Antarctica research station in the 1960s. And NASA’s two Van Allen Probes heard the chirping sounds from Earth's radiation belts at a closer distance than the newest observations.

The latest chorus waves were picked up by NASA's Magnetospheric Multiscale satellites. They were launched in 2015 to explore the Earth and the sun's magnetic fields.

Chorus waves have also been found near other planets including Jupiter and Saturn. The waves can produce high-energy electrons. That means they can threaten satellite communications.

The researchers said their findings suggest that chorus waves might be found everywhere in the magnetic fields of planets.

The study was released on January 22 in the scientific publication Nature.

I'm Anna Matteo.

Adithi Ramakrishnan reported this story for the Associated Press from New York. Anna Matteo adapted it for VOA Learning English.

Vocabulary

chorus - n. กลุ่มนักร้องประสานเสียง

significant - adj. สำคัญ, มีความหมาย

chirping - n. เสียงร้องจิ๊บๆ (ของนก), เสียงแหลมสั้นๆ

frequency - n. ความถี่

magnetic field - n. สนามแม่เหล็ก

antenna - n. เสาอากาศ

radiation belt - n. แถบรังสี (บริเวณรอบดาวเคราะห์ที่มีอนุภาคมีประจุไฟฟ้าสะสมอยู่)

Reading Comprehension Quiz

1. What is the primary subject of the study recently released by international researchers?

A. The impact of sound waves on human hearing in space.

B. Bursts of energy detected in an unexpected area of space.

C. The formation of new planets beyond our solar system.

D. Communication methods for future space missions.

2. Why do scientists refer to the energetic activity as 'chorus waves'?

A. Because they are always detected near a large group of stars.

B. Due to their ability to create musical notes in space.

C. Because they move at the same frequency as sounds humans can hear.

D. After the name of the scientist who first discovered them.

3. What do chorus waves sound like when converted into audio signals?

A. The sound of distant thunder.

B. Whistling winds.

C. Birds chirping.

D. A continuous hum.

4. What is the significant new finding of the latest research regarding chorus waves?

A. Their existence has just been discovered for the first time.

B. They are much louder than previously thought.

C. They exist at an unprecedented distance from a planet.

D. They can be artificially generated in laboratories.

5. According to the latest research, how far from Earth were the chorus waves detected?

A. 10,000 kilometers.

B. 50,000 kilometers.

C. 100,000 kilometers.

D. 500,000 kilometers.

6. What did study writer Chengming Liu state about chorus waves in space?

A. They are difficult to detect.

B. They are among the weakest waves in space.

C. They are one of the strongest and most significant waves.

D. They have no impact on Earth's environment.

7. What was Allison Jaynes's reaction to the recent study, as she was not involved in it?

A. She expressed skepticism about the findings.

B. She believed it confirmed existing theories about space physics.

C. She noted it opens up many new questions about possible physics in that area.

D. She criticized the research for being too narrow in scope.

8. In what specific area were the newfound chorus waves detected?

A. In the deep space beyond the solar system.

B. In an area where Earth's magnetic field is stretched out.

C. Within Earth's atmosphere, close to the surface.

D. Near the moon's gravitational field.

9. What new questions does the discovery of chorus waves in an unexpected area raise?

A. How to use chorus waves for interstellar communication.

B. The methods for converting chorus waves into electricity.

C. How the chirping waves form.

D. The exact temperature required for chorus waves to exist.

10. What is the current understanding among scientists regarding how chorus waves develop?

A. They are formed by solar flares.

B. They are a direct result of asteroid collisions.

C. Scientists are not sure, but Earth’s magnetic field may be involved.

D. They are caused by human-made satellites.

11. Which of the following have historically detected chorus waves?

A. Deep-sea submersibles.

B. Ancient astronomical observatories.

C. Radio antennas and NASA’s Van Allen Probes.

D. Specialized seismographs.

12. Which specific NASA satellites picked up the *latest* chorus waves mentioned in the article?

A. Voyager 1 and 2.

B. Hubble Space Telescope.

C. Magnetospheric Multiscale satellites.

D. James Webb Space Telescope.

13. What is one potential negative consequence of chorus waves mentioned in the text?

A. They can cause disruption to Earth's weather patterns.

B. They can threaten satellite communications.

C. They might accelerate climate change.

D. They can make space travel impossible for humans.

14. What broad implication do the researchers' findings suggest about chorus waves?

A. They are unique to Earth's magnetic field.

B. They are only found near gas giant planets.

C. They might be found everywhere in the magnetic fields of planets.

D. They are a rare phenomenon occurring only once every few centuries.

15. When was the study on chorus waves released in the scientific publication Nature?

A. December 15.

B. January 22.

C. February 1.

D. March 10.